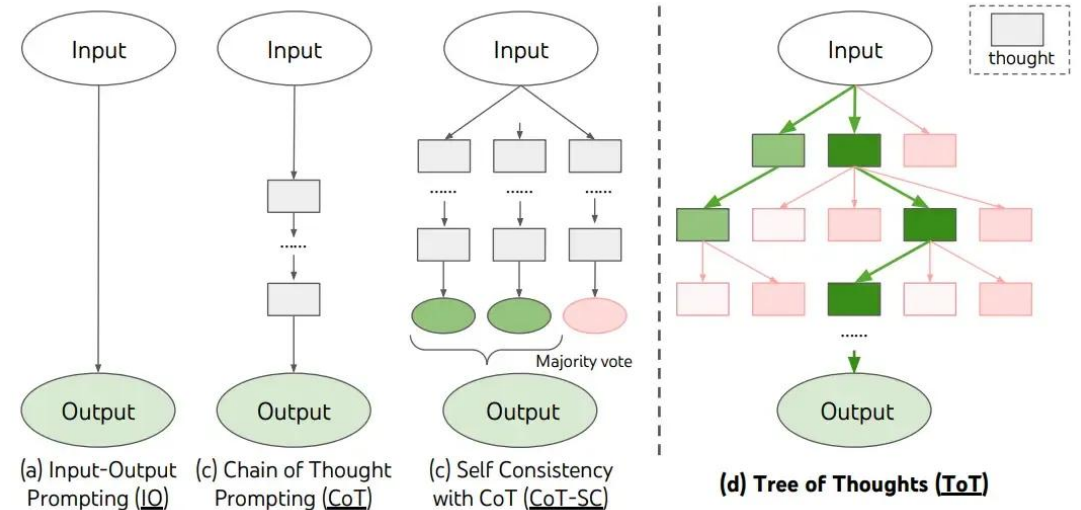


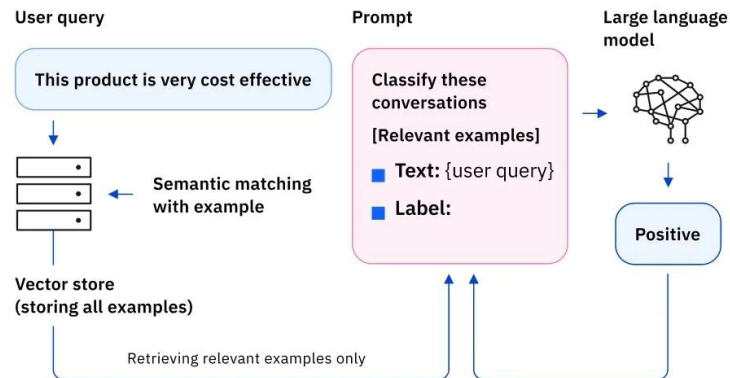
Chain-of-Thought (CoT): Step-by-Step Reasoning

- Chain-of-Thought is a prompting technique that encourages the model to break down complex problems into intermediate steps.
- By adding phrases like "Let's think step by step," we force the AI to solve the logic before reaching a final conclusion.
- This is essential for solving math word problems, symbolic logic, and multi-step reasoning where "Zero-shot" (direct) answering often fails.



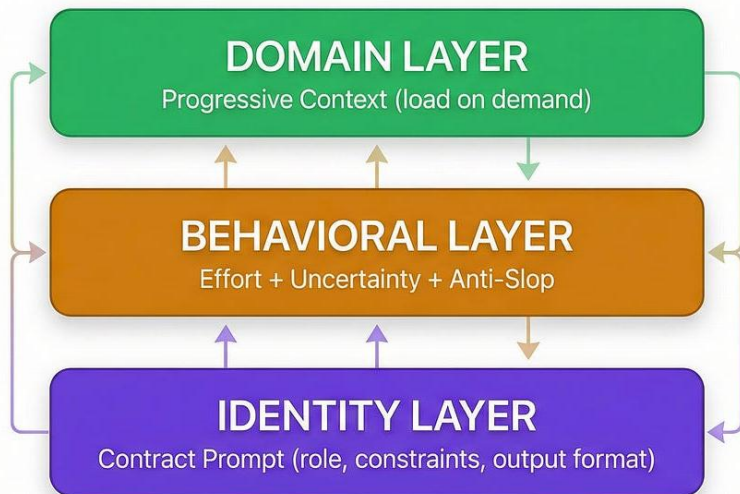
Few-Shot Prompting: Learning from Examples

- Instead of just giving a command, we provide the model with a few examples of the desired input-output behavior.
- **Zero-shot:** No examples given.
- **One-shot:** A single example provided.
- **Few-shot:** Multiple examples given to refine tone, format, and logic.
- This helps the model understand complex patterns or specialized formatting without fine-tuning.



System Prompts: Identity and Constraints

THE 3-LAYER SYSTEM PROMPT ARCHITECTURE



620 tokens total

- The System Prompt (or System Message) acts as the "master instruction" for the LLM.
- It sets the **Identity** (e.g., "You are a strict medical auditor") and **Constraints** (e.g., "Never use jargon" or "Keep responses under 50 words").
- It provides a consistent framework for how the model should interact with the user throughout the session.